

CSC1006 - Assignment 1

Q1 [10 pts]: For each of the following scientific tasks, state whether it is best framed as a Supervised, Unsupervised, Semi-supervised, or Reinforcement Learning problem. Justify your answer in one sentence.

A, Classifying galaxy morphologies (e.g., spiral, elliptical) from a large, fully labeled dataset of telescope images;

B, Grouping seismic sensor readings to identify novel patterns of underground activity without any pre-existing labels;

C, Training a model to control a robotic arm for precise sample handling in a lab, where it learns from rewards (successful grasp) and penalties (dropped sample);

D, Predicting molecular properties using a small set of experimentally measured molecules and a large database of unlabeled molecular structures.

A: Supervised (images are fully labeled, so the machine can just learn direct mapping image features to known classification)

B: Unsupervised (No pre-existing labels, so the goal is to find hidden patterns from data)

C: Reinforcement (Learn by interacting with environment and improve its performance by rewards and penalties)

D: Semi-supervised (The task combined small labeled dataset with large unlabeled dataset)

Q2 [10 pts]: What is the relation between Machine Learning (ML) and Deep Learning (DL)?

Deep learning is the technique to implement machine learning, which is inspired by brains. DL uses hierarchical representations learning on the raw data to find out features that ML can apply to find out the functions suitable for the problem.

Q3 [10 pts]: Which of ML and DL is more appropriate for the following respective tasks and explain why.

1), Facial recognition in images; DL (High dimensions/require hierarchical feature learning)

2), Predicting house prices based on a tabular dataset of 50 entries; ML (Dataset is very small and tabular)

3), Language translation; DL (it needs to analyze the structure and context where DL works better)

4), Identify different types of movies based on a dataset without the type labels.

ML (No labels → unsupervised, clustering)

Q4 [5 pts]: What are the two most common supervised learning tasks?

- Regression

- Classification

Q5 [10 pts]: What is the difference between training data and testing data in machine learning? why do you need testing data?

Training data: the machine uses them to find out patterns/rules which leads to the functions

Testing data: situations it has never faced before — checking if machine actually learned and evaluating its performance (done to check generalization and detect overfitting)

Q6 [10 pts]: You train two machine learning models on the same scientific dataset (predicting protein stability from sequence features)

Model 1: training error = 0.05, testing error = 0.08

Model 2: training error = 0.02, testing error = 0.15

Which model is likely overfitting? Explain your reasoning.

Model 2, since its training error is less than of the Model 1, but the testing error is more.

Q7 [10 pts]: Would you frame spam detection as a supervised learning or unsupervised learning problem? can you name one unsupervised learning problem?

Spam detection $\rightarrow$ supervised learning (discrete output yes/no from classification and the contents' relatability)

Unsupervised learning example: Clustering

Q8 [10 pts]: A linear model is trained to predict the yield (%)

of a chemical reaction based on three features: Temperature (in Celsius), Catalyst Amount (in mg), and Reaction time (in hours). The final model you learned is : $\text{Yield} = 15.0 + 0.8 \times \text{Temperature} + 2.5 \times \text{Catalyst_Amount} - 1.2 \times \text{Reaction_time}$

1). Interpret the coefficient for Reaction time. What does it signify and magnitude tell you?

-1.2 $\rightarrow$ every 1 hour increase in reaction time, the predicted yield decreases by 1.2 percentage points (assuming temperature and catalyst amount are constant)

2). If Temperature is measured in Kelvin instead of Celsius, how would you expect the coefficient 0.8 to change? (Hint : $\text{Kelvin} = \text{Celsius} + 273.15$). Would the model's predictions change? Justify.

$$\because T_c = T_k - 273.15 \rightarrow 0.8 \times T_c = 0.8 \times (T_k - 273.15) = 0.8 T_k - 218.52$$

$\rightarrow$ The coefficient of temperature remains, only intercept changes, hence the model's prediction does not change

Q9 [5 pts]: Write down the loss function formula for Linear Regression

$$L_{w,b} = \frac{1}{N} \sum_{i=1}^N [y^{(i)} - (w \cdot x^{(i)} + b)]^2$$

Q10 [10 pts]: You are given a toy dataset contains x (array of temperature) and y (array of density of one fluid), and you use a linear model hypotheses $\hat{y} = wx + b$ to model the physical relationship between the fluid density and temperature. The dataset are in following

$$\text{MSE} = \frac{1}{3} \sum_{i=1}^3 (y^{(i)} - (w x^{(i)} + b))^2 = \frac{1}{3} [(2.0 - (1.2 \cdot 1.0 + 0.5))^2 + (3.5 - (1.2 \cdot 2.0 + 0.5))^2 + (4.5 - (1.2 \cdot 3.0 + 0.5))^2]$$

x	y
1.0	2.0
2.0	3.5
3.0	4.5

$$= 0.203$$

1). Manually calculate the Mean Squared Error (MSE) loss for the entire dataset given initial parameters $w = 1.2$ and $b = 0.5$.

2). Compute the gradient (i.e., derivative) of the loss with respect to w (i.e., $\frac{\partial L}{\partial w}$ for the first data point ($x = 1.0$, $y = 2.0$)) using this initial (w, b) .

$$\frac{\partial L}{\partial w} = \frac{1}{3} \sum_{i=1}^3 2(y^{(i)} - (w \cdot x^{(i)} + b))(-x^{(i)}) = 2(2.0 - (1.2 \cdot 1.0 + 0.5))(-1.0) = -0.6$$

Q11 [10 pts]: Suppose you are given the following data set X,Y containing 20 records of time for X (in hours) and displacement of one car for Y (in km):

X = array([[1.19154063, 1.37434891, 1.17195486, 0.16152634, 0.79415049, 1.72423116, 1.9907262, 0.2851336, 1.78185502, 1.2749914, 1.35433286, 0.37991704, 1.86381926, 0.14110919, 0.4916442, 0.08485453, 1.40743255, 1.8189095, 0.79861128, 0.91450109]])

Y = array([[7.25283192, 7.4995625, 8.84322609, 3.94706203, 6.65112617, 8.23952238, 10.00317942, 3.19168625, 10.03715158, 7.27914416, 8.66561184, 3.73632541, 9.58035831, 3.99475623, 5.84385689, 3.19935995, 8.69952762, 8.48467547, 6.81720262, 6.05033212]])

Perform a **Linear Regression** on this dataset, what will the velocity (i.e., slope w in linear regression) and starting point displacement (i.e., intercept b in linear regression) be?

$$w \approx 3.41$$

$$b \approx 3.32$$

A1_Q1.py
A1_Q2.py
A1_Q3.py
A1_Q4.py
A1_Q5.py
A1_Q6.py
A2_Q1.py
A2_Q2.py
A2_Q3.py
A2_Q4.py
A2_Q5.py
A3_Q1.py
A3_Q2.py
A3_Q3.py
A4_Q1.py
A4_Q2.py
A4_Q3.py
bru.hpy
d.py
week3.py
week4.py
week5.2.py

```
1 import numpy as np
2 def linear_regression(x, y): 1usage
3     # Convert to numpy arrays (in case lists are given)
4     x = np.array(x)
5     y = np.array(y)
6
7     # Compute means
8     x_mean = np.mean(x)
9     y_mean = np.mean(y)
10
11     # Compute slope (w)
12     numerator = np.sum((x - x_mean) * (y - y_mean))
13     denominator = np.sum((x - x_mean) ** 2)
14     w = numerator / denominator
15     # Compute intercept (b)
16     b = y_mean - w * x_mean
17     return w, b
18
19 x_values = [1.19154063, 1.37434891, 1.17195486, 0.16152634, 0.79415049, 1.72423116,
20 1.9907262, 0.2851336, 1.78185502, 1.2749914, 1.35433286, 0.37991704, 1.86381926, 0.14110919,
21 0.4916442, 0.08485453, 1.40743255, 1.8189095, 0.79861128, 0.91450109]
22 y_values = [7.25283192, 7.4995625, 8.84322609, 3.94706203, 6.65112617, 8.23952238,
23 10.00317942, 3.19168625, 10.03715158, 7.27914416, 8.66561184, 3.73632541, 9.58035831,
24 3.99475623, 5.84385689, 3.19935995, 8.69952762, 8.48467547, 6.81720262, 6.05033212]
25 w, b = linear_regression(x_values, y_values)
26 print("Slope (w):", w)
27 print("Intercept (b):", b)
```

C:\Users\Lenovo\PycharmProjects\PythonProject1\.venv\Scripts\python.exe C:\Users\Lenovo\PycharmProjects\PythonProject1\d.py

Slope (w): 3.4100574559610317

Intercept (b): 3.31931148942666